

9.1. Determine the semi-minor and semi-major axes as well as the focal points of the ellipse $9x^2 + 25y^2 - 225 = 0$. Moreover, draw this ellipse.

9.2. Determine the position of the line $\ell : 2x + y - 10 = 0$ relative to the ellipse $\mathcal{E} : \frac{x^2}{9} + \frac{y^2}{4} - 1 = 0$.

9.3. Determine an equation of a line which is orthogonal to $\ell : 2x - 2y - 13 = 0$ and tangent to the ellipse $\mathcal{E} : x^2 + 4y^2 - 20 = 0$.

9.4. Determine the common tangents to the ellipses

$$\frac{x^2}{45} + \frac{y^2}{9} = 1 \quad \text{and} \quad \frac{x^2}{9} + \frac{y^2}{18} = 1.$$

9.5. Consider the ellipse $\mathcal{E} : \frac{x^2}{4} + y^2 - 1 = 0$ with focal points F_1 and F_2 . Determine the points M , situated on the ellipse, for which the angle $\angle F_1MF_2$ is a right angle.

9.6. Using a rotation of the coordinate system, find the equation of an ellipse centered at the origin, with focal points on the line $x = y$ and having semi-major axes equal to 4 and the distance between the focal points equal to $2\sqrt{3}$.

9.7. Consider the family of ellipses $\mathcal{E}_a : \frac{x^2}{a^2} + \frac{y^2}{16} = 1$. For what value $a \in \mathbb{R}$ is $\mathcal{E}_a$ tangent to the line $\ell : x - y + 5 = 0$?

9.8. Consider the family of lines $\ell_c : \sqrt{5}x - y + c = 0$. For what values $c \in \mathbb{R}$ is ℓ_c tangent to the ellipse $\mathcal{E} : x^2 + \frac{y^2}{4} = 1$?

9.9. A *diameter* of an ellipse is the line segment determined by the intersection points of the ellipse with a line passing through the center of the ellipse. Show that the tangent lines to an ellipse at the endpoints of a diameter are parallel.